

Math 196 Problem Set 6

Due May 7, at 10:00 PM

Problem 1. For each of the following vectors v , compute the length $|v|$.

$$\begin{bmatrix} 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1/3 \\ 1 \end{bmatrix}$$

Problem 2. For each pair of vectors, compute $\cos(\theta)$, where θ is the angle between them.

(a) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

Problem 3. Sketch approximate pictures of pairs of vectors $v, w \in \mathbb{R}^2$ satisfying the following conditions.

- (a) $v \cdot w < 0$
- (b) $v \cdot w = 0$
- (c) $|v|$ and $|w|$ are large, and $v \cdot w$ is small and positive.
- (d) $|v|$ is small, and $v \cdot w$ is very negative.

Problem 4. Let $W \subset \mathbb{R}^n$ be a linear subspace. In class, we defined $W^\perp$ as the set of vectors

$$\{v \in \mathbb{R}^n \mid v \cdot w = 0 \text{ for all } w \in W\}.$$

- (a) Prove from the definition that $W^\perp$ is a linear subspace of $\mathbb{R}^n$.
- (b) Prove that $W \cap W^\perp = \{0\}$, by showing that if $w \in W$ and $w \in W^\perp$, then $w = 0$.
- (c) Suppose that $v_1 + v_2 = v'_1 + v'_2$, where $v_1, v'_1 \in W$ and $v_2, v'_2 \in W^\perp$. Show that $v_1 = v'_1$ and $v_2 = v'_2$. That is, if a vector v can be decomposed as $v_1 + v_2$ for $v_1 \in W, v_2 \in W^\perp$, then this decomposition is unique.¹

Problem 5. Apply the Gram-Schmidt process to the following lists of vectors to produce orthonormal sets.

(a) $\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \\ -1 \end{bmatrix}$

¹Hint: Consider $v_1 - v'_1$ and $v_2 - v'_2$.

(c) $\begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$

Problem 6. Find the matrices representing orthogonal projection onto the following linear subspaces.

(a) $\text{span} \left\{ \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 6 \\ 4 \end{bmatrix} \right\}$

(b) The solution set to the equation $x + y + z = 0$.

Problem 7. For each subspace, find a basis (not necessarily orthonormal) of its orthogonal complement.

(a) The solution set of the system of equations:

$$\begin{cases} x_1 + x_2 + x_3 = 0 \\ 2x_2 - x_3 - x_4 = 0 \\ x_1 + 3x_2 - x_4 = 0 \end{cases}$$

(b) $\text{im} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 3 \\ 1 & 1 \end{bmatrix}$

Problem 8. Consider the subspace

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3$$

(a) Find a basis (not necessarily orthonormal) for $W^\perp$.

(b) Argue that together with the vector spanning W , these vectors form a basis $\mathfrak{B}$ for $\mathbb{R}^3$, and find the matrix B for orthogonal projection onto W with respect this basis.

(c) Find the matrix A for orthogonal projection onto W with respect to the standard basis, and write down an equation of the form

$$B = S^{-1}AS$$

for an appropriately chosen S , showing that A and B are similar. No need to explicitly compute S^{-1} or the matrix products here, the equation should come from the fact that B and A represent the same transformation with respect to different bases.